

MANAV PATEL

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EDUCATION

MERCER UNIVERSITY

Macon, Georgia

- Bachelor of Science in Computer Science
- Minor in Information Science and Technology

EXPERIENCE

Terma Group

Warner Robins, Georgia

Software Engineer Intern

May 2025 – August 2025

- Designed and developed a .NET-based virtual hardware simulator with interactive screens and buttons, enabling testing and interaction with avionics controllers via UDP networking.
- Reverse engineered undocumented communication protocols by analyzing controller traffic with Wireshark and writing Lua scripts to parse packets and decode message structures.
- Debugged complex data mismatches between provided documentation and actual controller output, ensuring accurate simulation and communication.
- Built responsive UI components in C#, WPF, and XAML under the .NET framework, applying MVVM for maintainable and scalable design.
- Pair programmed with mentor to implement and refine server and client testing code, driving overall system reliability, and integrated UI with system logic while verifying behavior directly against physical hardware.
- Utilized the Atlassian suite, including Jira for Agile sprint planning, Bitbucket for version control and code reviews, and Confluence for project documentation, and acted as a consultant for another intern's project, leveraging Docker and pipeline testing expertise to guide development and improve workflow.
- Gained hands-on experience with aerospace-grade software workflows and the complete software development lifecycle (SDLC).

PROJECTS

Fitness Tracker App – Self-initiated

- Developing a fitness tracking app using Swift for iOS
- Focused on features such as step counting, workout logging, and user activity tracking
- Exploring integration with Apple HealthKit for seamless health data syncing
- Utilizing a Gantt chart with aspects of agile framework to manage project timelines, tasks, and milestones, improving time management and planning skills

Five Card Stud Poker, Mercer University

Macon, Georgia

- Developed a comprehensive Five Card Stud Poker simulation program at Mercer University
- Implemented functionality for hand creation, hand ranking, and tie-breaking using various data structures
- Proficiently coded in C++, C#, Java, Python, Fortran, Go, Julia, Lisp, Pearl, and Rust
- Demonstrated versatility and adaptability across a diverse range of programming languages
- Showcased problem solving skills and expertise in software development through project execution

NSF Rhesus Monkey Research Database, Mercer University

Macon, Georgia

NSF Research collaborator

- Developed and managed a database for NSF research on rhesus monkeys, utilizing SQL for efficient data retrieval and storage
- Extracted and processed data from Excel sheets, ensuring accurate and reliable data integration
- Wrote complex SQL queries to perform key data comparisons by supporting research conclusions and drawing insights from the data
- Contributed to automating data handling and reporting for scientific publications, enabling more efficient analysis

Advanced Algorithms and Robotics, Stratford Academy

Macon, Georgia

Robotics Lead/Tester

- Led the end-to-end software and hardware development of a groundbreaking web-controlled robot, enabling remote students affected by COVID-19 to virtually navigate school corridors and engage with peers through video calls
- Coded a website using HTML and CSS with its infrastructure based in Java
- Utilized CAD software to design the shell and base of the machine which were then 3D printed to exact specifications
- Orchestrated a dual-system setup, harnessing ESP8266 for hardware operations and Raspberry Pi for seamless video interface and processing, ensuring optimal functionality and performance

SKILLS/INTERESTS

Programming Languages: C++, C#, .NET, XML, Swift, Java, Python, SQL, Lua, Rust

Software: Microsoft Office, Atlassian suite, Arduino IDE, Vim, Visual Studio Code, GIT, Xcode, Wireshark, Firebase

Languages: Native proficiency in English and Gujarati, Beginner proficiency in Hindi